

Advanced Power BI Techniques

This workbook contains some advanced techniques to pre-process data and create new calculated columns.

We are starting from the table below and we want to create a chart that shows the top 3 employees for each project (the employees that worked the most on that project).

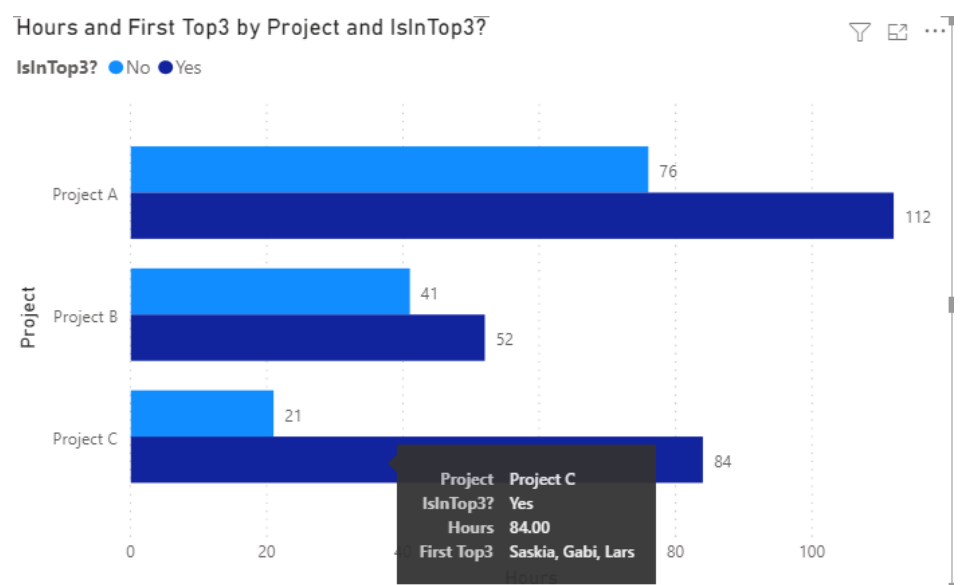
	A ^B _C Column1	A ^B _C Column2	1 ² ₃ Peter	1 ² ₃ Mickey	1 ² ₃ Lars	1 ² ₃ Kim
1	Week 1	Project A	14	15	28	4
2	null	Project B	6	null	null	null
3	null	Project C	null	2	9	3
4	Week 2	Project A	13	4	16	11
5	null	Project B	null	null	null	5
6	null	Project C	null	10	14	2
7	null	null	33	31	67	25

The table contains the list of projects (A, B and C), and how many hours each employee (Peter, Mickey, etc..) has worked on those projects in week 1 or week 2.

In the final chart below, we want to display the number of hours spent by the employees on each project, but we want to split them into two classes: the total number of hours spent by the top 3 employees (the three employees that worked the most on that project) and the number of hours everyone else spent on that project.

For example, for Project C the top 3 employees (Saskia, Gabi and Lars) worked 84 hours in total on that project, while the other employees worked 21 hours in total.

The most challenging part is to find a way to sort our data and create a list of the top 3 employees for each project.



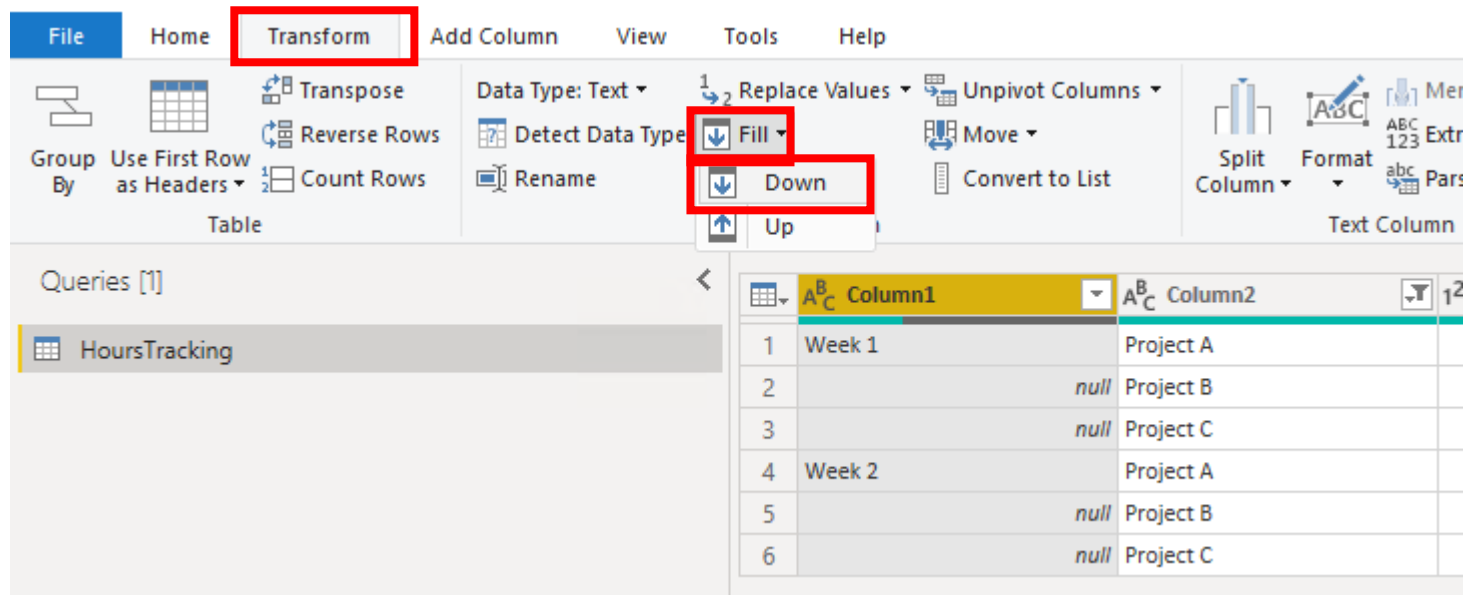
We also need to use the Unpivot tool to have a column with the employees' names and the hours worked.

The tasks below use the "Working Hours" dataset available on Blackboard. Download the file before starting the tasks.

Import the "Working Hours" dataset then use Transform Data to remove nulls from Column 2.

We need to transform the table so that you have the following columns: Project, Week, Employee, Hours Worked.

The first step is to fill in the nulls below "Week 1" and "Week 2" with the correct values. To do this, Select column 1 then click on Transform and Fill-> Down



File Home **Transform** Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [1]

HoursTracking

Data Type: Text

Replace Values

Unpivot Columns

Split Column

Format

Text Column

Fill

Down

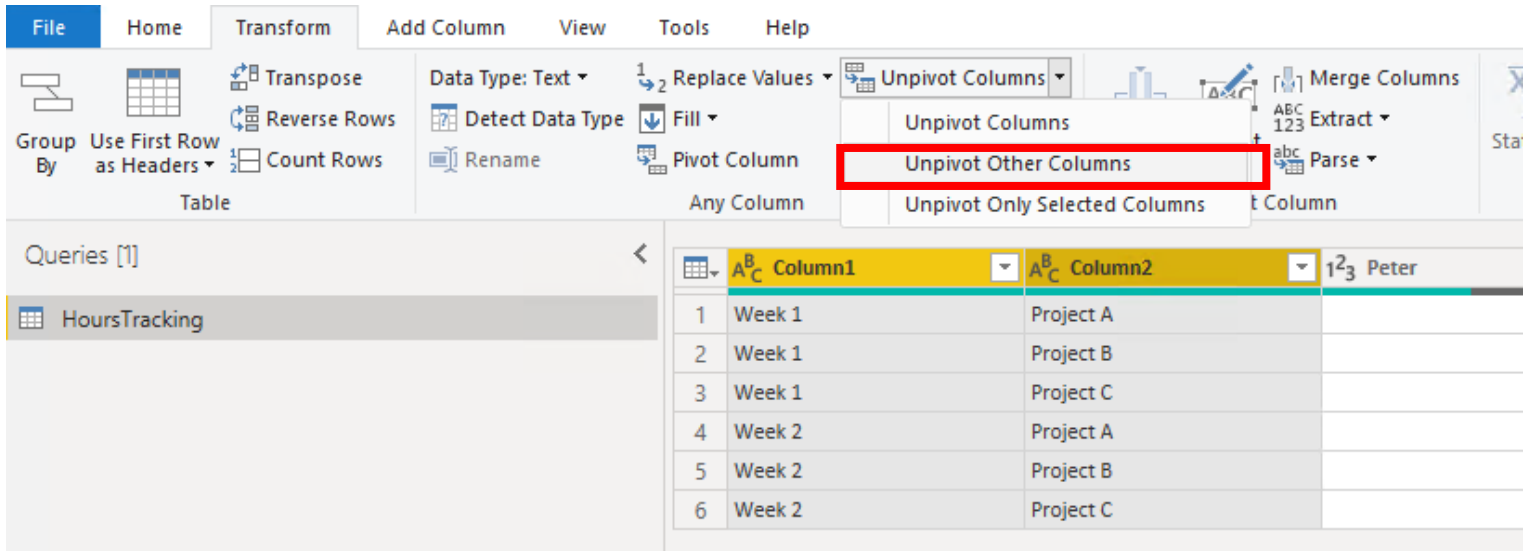
Up

	Column1	Column2
1	Week 1	Project A
2		Project B
3		Project C
4	Week 2	Project A
5		Project B
6		Project C

The new column is shown below.

	Column1
1	Week 1
2	Week 1
3	Week 1
4	Week 2
5	Week 2
6	Week 2

To have a column with just the employees' names, we need to unpivot the table. To do this, select Column 1 and Columns 2 by clicking on them (keep pressing the shift-key on your keyboard for multiple selections), then click on Transform -> Unpivot Columns and Unpivot Other Columns (as shown below).

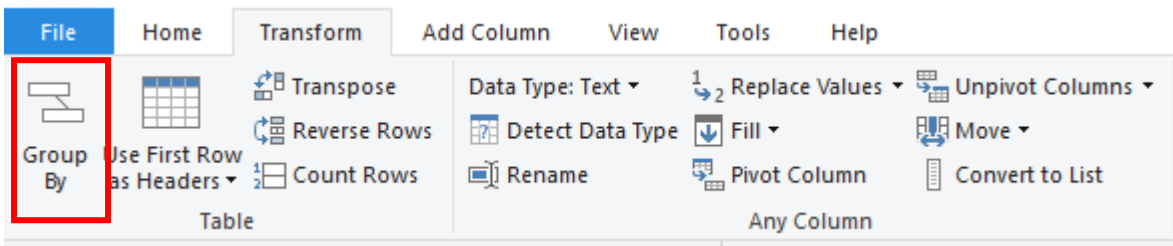


	Week	Project	Employee
1	Week 1	Project A	Peter
2	Week 1	Project B	Mickey
3	Week 1	Project C	Lars
4	Week 2	Project A	Kim
5	Week 2	Project B	Marc
6	Week 2	Project C	Beth

The new table has now an Attribute column (employees' names) and a Value column (Hours worked). Rename all the columns as shown below.

	Week	Project	Employee	Hours
1	Week 1	Project A	Peter	14
2	Week 1	Project A	Mickey	15
3	Week 1	Project A	Lars	28
4	Week 1	Project A	Kim	4
5	Week 1	Project A	Marc	24
6	Week 1	Project A	Beth	7

In order to check the top employees for each project, we need to first apply some sorting and grouping. Click on Group under Transform.



Add the details below in the Group by window

Group By

Select Advanced

Specify the columns to group by and one or more outputs.

☐ Basic ☒ Advanced

Project

Employee

Add grouping

New column name

Hours

Add aggregation

Operation

Sum

Column

Hours

We want to group the values in the table by Project first and then by Employee (Click Add grouping to add the second level for Employee)

For each project and each employee who worked on that project, you want to work out the Sum (Operation) of the Hours (Column) and we are saving this as a new column named "Hours" (New column name)

OK

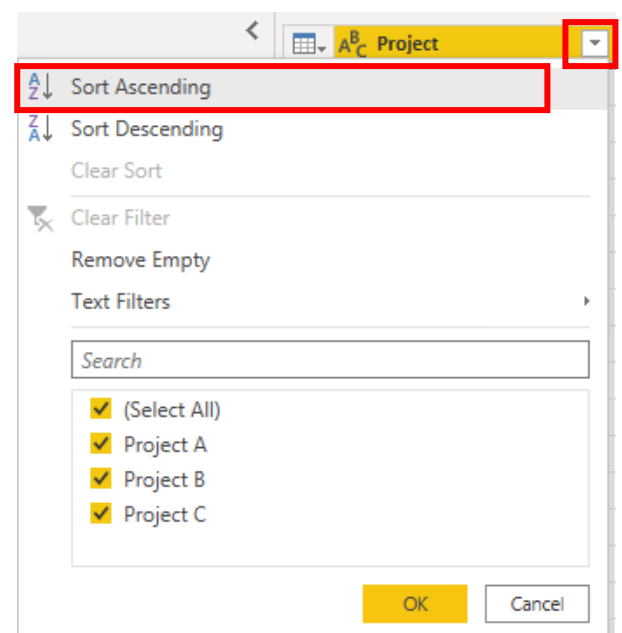
Cancel

Click OK to see the table below, which provides the total number of hours each employee has worked on each project (either A, B or C).

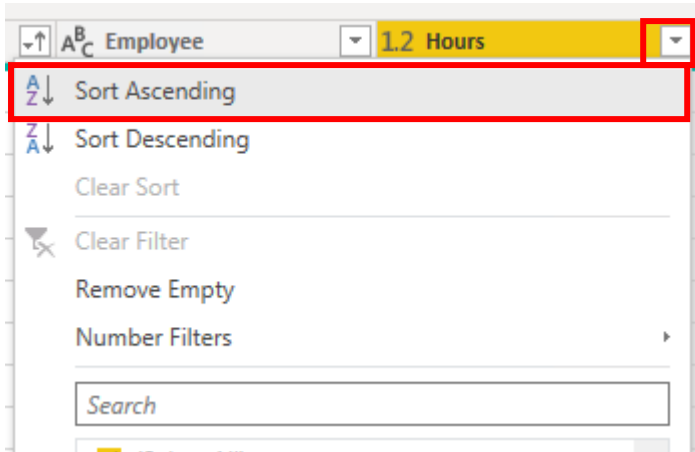
	A ^B _C Project	A ^B _C Employee	1.2 Hours
1	Project A	Peter	27
2	Project A	Mickey	19
3	Project A	Lars	44
4	Project A	Kim	15
5	Project A	Marc	41
6	Project A	Beth	25
7	Project A	Michael	8
8	Project A	Stefanie	9
9	Project B	Peter	6
10	Project B	Marc	8
11	Project B	Beth	3
12	Project B	Michael	7
13	Project B	Stefanie	16
14	Project B	Gabi	6
15	Project B	Saskia	6

Sorting

Now, we need to apply the correct sorting to have the top employees for each project at the top of the corresponding column. To do this, click on the drop-down arrow in the Project column and select sort ascending.



Repeat this for the Hours column, but this time select “Sort Descending”, since you want the biggest numbers at top of the column.



Using the resulting table on the right, it is now easy to identify the top-3 employees (the ones who worked the highest number of hours) for each project.

	Project	Employee	Hours
1	Project A	Lars	44
2	Project A	Marc	41
3	Project A	Peter	27
4	Project A	Beth	25
5	Project A	Mickey	19
6	Project A	Kim	15
7	Project A	Stefanie	9
8	Project A	Michael	8
9	Project B	Wesley	22
10	Project B	Stefanie	16
11	Project B	Jim	14
12	Project B	Marc	8
13	Project B	Michael	7
14	Project B	Saskia	6
15	Project B	Gabi	6
16	Project B	Peter	6
17	Project B	Kim	5
18	Project B	Beth	3
19	Project C	Saskia	33
20	Project C	Gabi	28
21	Project C	Lars	23
22	Project C	Mickey	12
23	Project C	Kim	5
24	Project C	Marc	4

Now that we have the correct sorting applied, we are going to group once more. Click on Transform -> Group By and fill in the details as shown below.

✕

Group By

Specify the columns to group by and one or more outputs.

☐ Basic
 ☒ Advanced

Project ▼

We want to group by project

Add grouping

New column name

Data

Add aggregation

Operation

All Rows ▼

Column

OK

Cancel

And we want to keep All Rows (Operation), but this time the New Column name is Data.

Let's see what this does. Click Ok.

	Project	Data
1	Project A	Table
2	Project B	Table
3	Project C	Table

You now have a small table with only the three project and the rest of the data “grouped” in the Data column (which contains 3 tables). If you click on the first Table you’ll see the table below.

	A ^B _C Project	A ^B _C Employee	1.2 Hours
1	Project A	Lars	44
2	Project A	Marc	41
3	Project A	Peter	27
4	Project A	Beth	25
5	Project A	Mickey	19
6	Project A	Kim	15
7	Project A	Stefanie	9
8	Project A	Michael	8

Query Settings

PROPERTIES

Name: HoursTracking

[All Properties](#)

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Filtered Rows
- Filled Down
- Unpivoted Other Columns
- Renamed Columns
- Grouped Rows
- Sorted Rows
- Grouped Rows1
- Project A

This table collects all the rows related to project A. Click the cross next to Project A to go back to the main Table.

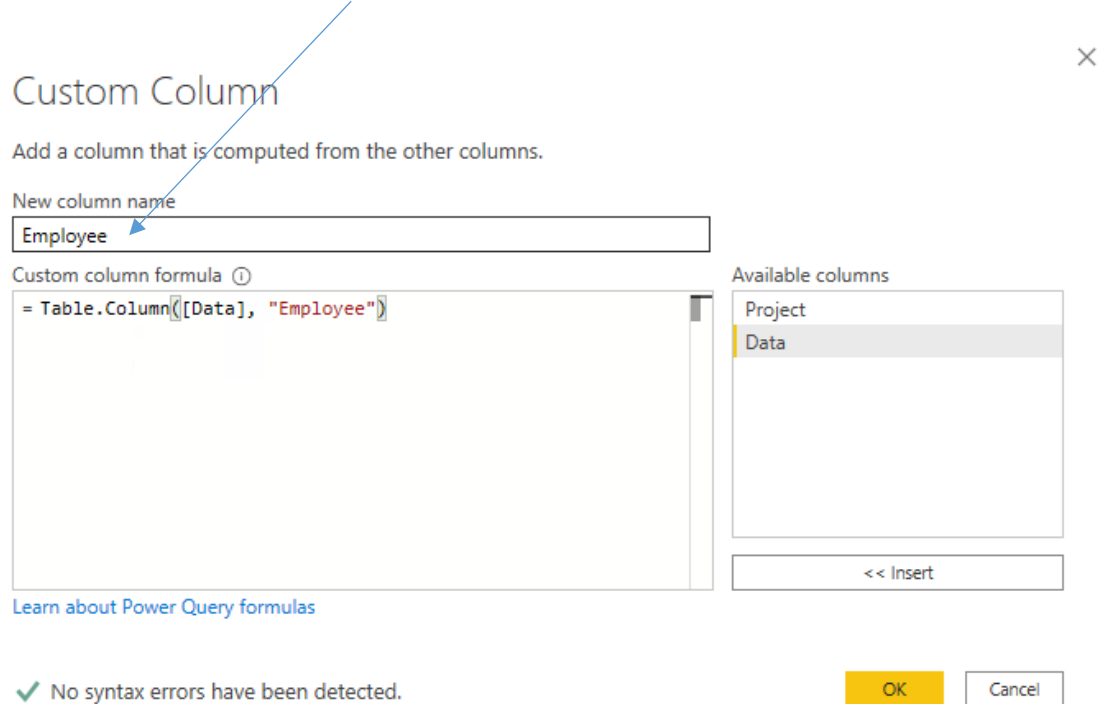
We now have a table that contains three smaller tables (one for each project). We need to extract only the List of employees from each one of these Data smaller tables (remember our main goal is to select the top 3 employees for each project).

	A ^B _C Project	Data
1	Project A	Table
2	Project B	Table
3	Project C	Table

To create a column with the list of the Employees, go to Add Column and select Custom Column.

We are going to call the new column “Employee” and use the formula shown below to select only the list of employees.

The Table.Column function uses the Data table to extract the “Employee” column and save the result in the New Column Name – Employee



Custom Column

Add a column that is computed from the other columns.

New column name
Employee

Custom column formula ⓘ
= Table.Column([Data], "Employee")

Available columns
Project
Data

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

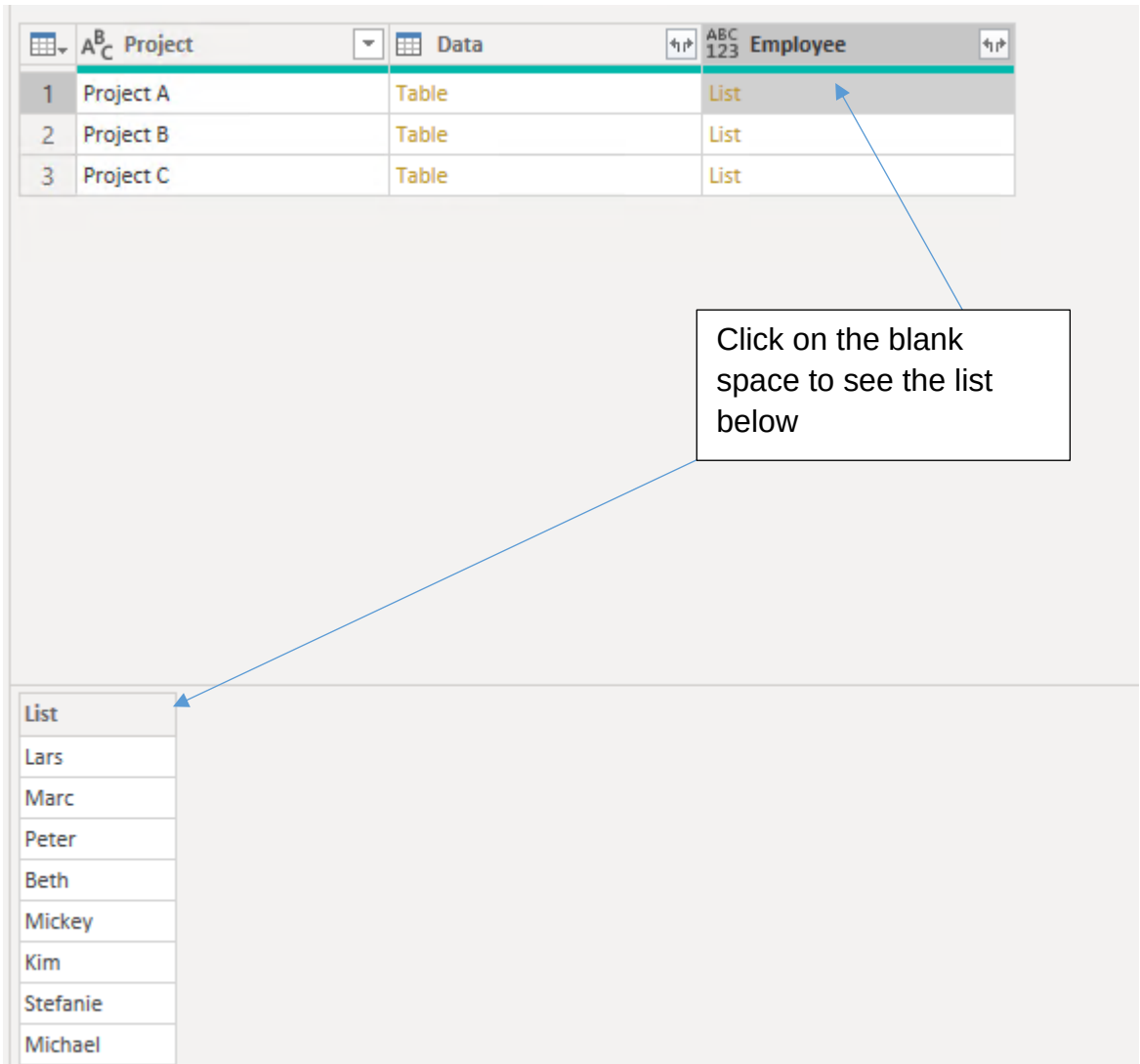
OK Cancel

The new table is shown below.

	Project	Data	Employee
1	Project A	Table	List
2	Project B	Table	List
3	Project C	Table	List

As you can see, there is now a List for each Project.

To check the names on that list, click on the blank space next to List. This will display the list of employees who worked on that project.



The screenshot shows a data application interface. At the top, there are two tabs: 'Project' and 'Employee'. The 'Project' tab is active, displaying a table with three rows: Project A, Project B, and Project C. The 'Employee' tab is also visible, showing a list of names. A callout box points to the 'List' column header in the 'Employee' tab, indicating that clicking on the blank space next to 'List' will display the list of employees who worked on that project.

Project	Table	List
1	Project A	List
2	Project B	List
3	Project C	List

Click on the blank space to see the list below

List
Lars
Marc
Peter
Beth
Mickey
Kim
Stefanie
Michael

Now that we have the ordered list of the employees, we can select the first three rows from each list.

To do this, we need to create another custom column (Top3), with only the first three names. Click on Add Column -> Custom Column and use the M language formula shown below.

The name of the new column is "Top3". We are using the List.FirstN() function to select the first 3 rows from the [Employee] column.

Custom Column

Add a column that is computed from the other columns.

New column name

Top3

Custom column formula ①

= List.FirstN([Employee], 3)

Available columns

Project
Data
Employee

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

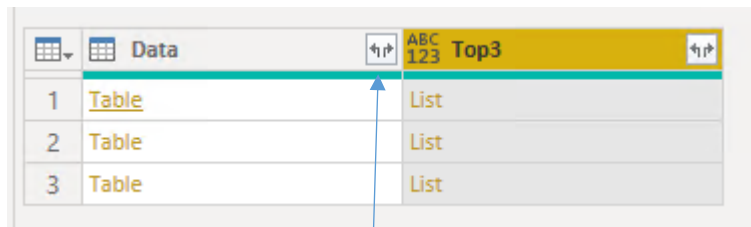
Cancel

Click Ok, to see the new table below. Click on the blank space to see the top3 employees for each project.

	Project	Data	Employee	Top3
1	Project A	Table	List	List
2	Project B	Table	List	List
3	Project C	Table	List	List

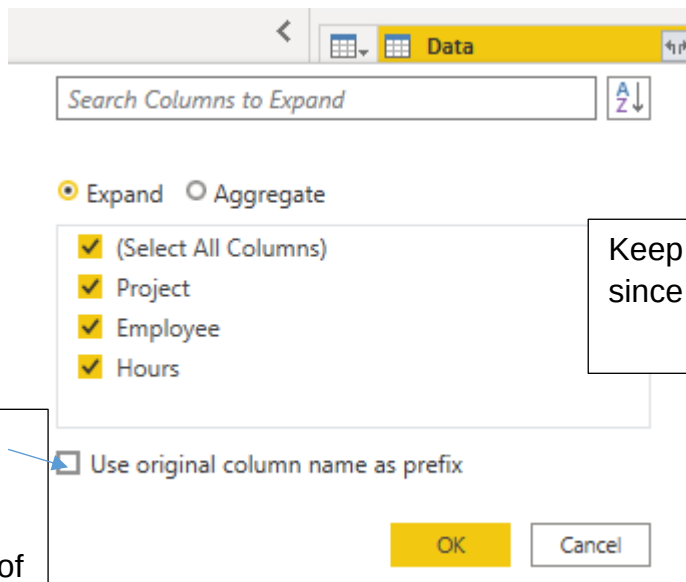
List
Lars
Marc
Peter

Now that we finally have our Top3 column, we can remove the columns that we don't need. Remove the Project and Employee Columns (they are both in the Data column as nested table, so we can retrieve them from there in the next step).



	Data	Top3
1	Table	List
2	Table	List
3	Table	List

Then click on the arrows in the Data column's header to expand that table.



Search Columns to Expand

☒ Expand ☐ Aggregate

- ☒ (Select All Columns)
- ☒ Project
- ☒ Employee
- ☒ Hours

☐ Use original column name as prefix

OK Cancel

Keep all the Columns Selected since you want to expand them all

Untick this box, so that you will just see the original names of the columns without the Data. prefix

We are now back to the original table, but we finally have a Top3 column with the list of the Top3 employees for each project.

	ABC Project	ABC Employee	1.2 Hours	ABC 123 Top3
1	Project A	Lars	44	List
2	Project A	Marc	41	List
3	Project A	Peter	27	List
4	Project A	Beth	25	List
5	Project A	Mickey	19	List
6	Project A	Kim	15	List
7	Project A	Stefanie	9	List
8	Project A	Michael	8	List
9	Project B	Wesley	22	List
10	Project B	Stefanie	16	List
11	Project B	Jim	14	List
12	Project B	Marc	8	List
13	Project B	Michael	7	List
14	Project B	Gabi	6	List
15	Project B	Saskia	6	List
16	Project B	Peter	6	List
17	Project B	Kim	5	List
18	Project B	Beth	3	List

To visualise the names of the employees in the table, click on the arrows to Expand the Top3 column and select Extract Values.



We would like to have the names separate by a comma, with a space after that.
To do this, select –Custom– and type , and space as shown below.,

Extract values from list

Select a delimiter to use for concatenating list values

--Custom--

,

☐ Concatenate using special characters

Insert special character

OK

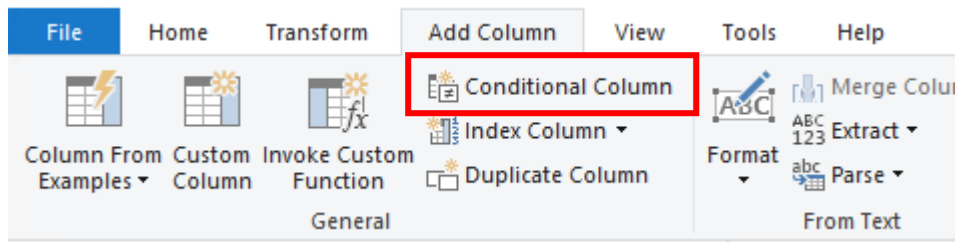
Cancel

The final table is shown below. For each Project, we have the list of the top3 employees.

	A_C^B Project	A_C^B Employee	1.2 Hours	A_C^B Top3
1	Project A	Lars	44	Lars, Marc, Peter
2	Project A	Marc	41	Lars, Marc, Peter
3	Project A	Peter	27	Lars, Marc, Peter
4	Project A	Beth	25	Lars, Marc, Peter
5	Project A	Mickey	19	Lars, Marc, Peter
6	Project A	Kim	15	Lars, Marc, Peter
7	Project A	Stefanie	9	Lars, Marc, Peter
8	Project A	Michael	8	Lars, Marc, Peter
9	Project B	Wesley	22	Wesley, Stefanie, Jim
10	Project B	Stefanie	16	Wesley, Stefanie, Jim
11	Project B	Jim	14	Wesley, Stefanie, Jim
12	Project B	Marc	8	Wesley, Stefanie, Jim
13	Project B	Michael	7	Wesley, Stefanie, Jim
14	Project B	Gabi	6	Wesley, Stefanie, Jim
15	Project B	Saskia	6	Wesley, Stefanie, Jim

Before starting with the report, we need an extra column that says “yes” or “no” to check if that employee is in the top3 employees for that specific column.

To do this, let’s add a **conditional column** and fill in the options as shown below. You can find the conditional column button in the Add Column ribbon.



The new “IsInTop3” column will display “yes” if the “Top3” column contains the Employee name, “no” otherwise.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

IsInTop3?

	Column Name	Operator	Value		Output
If	Top3	contains	 Employee	Then	ABC 123 Yes

Add Clause

Else

ABC 123

No

Make sure you select the table symbol here

OK

Cancel

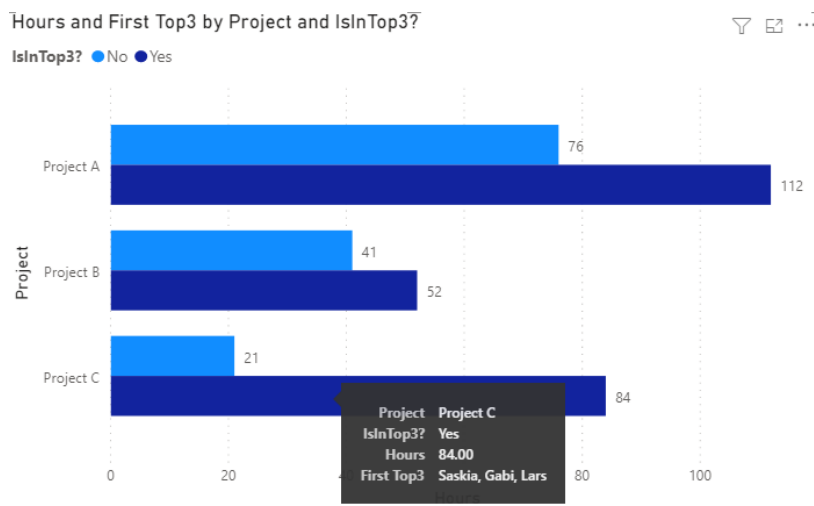
The final table is shown below. The new column contains Yes/No as defined in the formula. For example, Lars is in the top3 for Project A, but Beth is not.

	ABC Project	ABC Employee	1.2 Hours	ABC Top3	ABC 123 IsInTop3?
1	Project A	Lars		44 Lars, Marc, Peter	Yes
2	Project A	Marc		41 Lars, Marc, Peter	Yes
3	Project A	Peter		27 Lars, Marc, Peter	Yes
4	Project A	Beth		25 Lars, Marc, Peter	No
5	Project A	Mickey		19 Lars, Marc, Peter	No
6	Project A	Kim		15 Lars, Marc, Peter	No
7	Project A	Stefanie		9 Lars, Marc, Peter	No
8	Project A	Michael		8 Lars, Marc, Peter	No
9	Project B	Wesley		22 Wesley, Stefanie, Jim	Yes
10	Project B	Stefanie		16 Wesley, Stefanie, Jim	Yes

We are now ready to create the final report. Before doing so, click Close&Apply to apply these changes.

Your turn: creating the Final Visual.

Now that you have all the data in place, it's your turn to create the chart below.



You need a clustered bar chart with project on the axis, IsInTop3 on the legend, hours worked on values and on the tooltip Top3.

How would you add the data labels on each bar?
What about the extra tooltip that shows the FirstTop3?

Can you think of a way of sorting the chart like I did? With the smallest values at the top for each project?

And the Projects' names in order (A, B and C)?

Watch this video to check my solution: [link to video](#).

You should consider using these advanced strategies in your ICA for data pre-processing or creating new measures.